

EwE linked stanza recruitment - a brief users' guide



Version 1, 25 January 2022
Compatible with Ecopath with Ecosim version 6.7 α 1
Jeroen Steenbeek
Ecopath International Initiative, Spain



**Ecopath
International
Initiative**

Contents

Scope and purpose	2
Acknowledgements	2
Summary	3
Computations	3
Configuring linked stanza recruitment	4
References	5

Scope and purpose

This document provides usage instructions of linked stanza recruitment in the Ecopath with Ecosim food web modelling approach (Christensen and Walters, 2004; Heymans et al., 2016). If you have any questions, find bugs, or see ways to expand on this functionality please contact us at ewedevteam@gmail.com.

This document was last updated January 2022 to EwE version 6.7.0.0 Alpha prerelease.

Acknowledgements

Linked stanza recruitment was implemented by Jeroen Steenbeek after an idea by Villy Christensen under project [*EISA: Ecology and management of the invasive snow crab*](#).



Summary

Occasionally, ecosystem modellers run into issues where gender-split populations need explicit modelling. This can occur when empirical data is only available for male or female specimens, or when males and females of key species have markedly different behaviours. We have made a small modification to the Ecopath with Ecosim (EwE) multi-stanza model (Christensen and Walters, 2004), that allows linking recruitment to ensure proportional spawning between two multi-stanza groups, and thus connecting two gender-split populations at their foundation.

Computations

At the first Ecosim and Ecospace time steps, a base scalar is calculated:

$$a_i = \frac{N_{Age1_i}}{N_{Age1_j}} \quad \text{Eq. 1}$$

where a is the scalar, i is a multi-stanza group whose recruitment is driven by multi-stanza group j , and N_{Age1} is the number of individuals at age 1, e.g., the number of recruits, for multi-stanza groups i and j . This base scalar serves to maintain the ratio between the number of gender-split recruits during the simulations.

For every consecutive time step, the number of multi-stanza group i recruits is then set as follows:

$$N_{Age1_i} = a_i \times N_{Age1_j} \quad \text{Eq. 2}$$

Changes to the EwE computations were minimal; the most challenging code changes were made in the Ecospace IBM model (Walters et al., 2010), where parallel computations was reorganized to ensure that multi-stanza recruitment is calculated in the correct order. Ecospace does not yet calculate recruitment success from spatial overlap of the gender-split stanza; this could be a future refinement if needed.

An example of linked recruitment is shown in Figure 1. Here, Ecosim is executed twice for a model that contains separate multi-stanza groups for male and female snow crab. Both multi-stanza groups consist of two life stages each: juveniles and adults. Only the male snow crabs are fished. The model contains no change in environment and fishing other than intensifying effort of the snow crab fishery. In the left panel the multi-stanza recruitment is not linked, in the right panel female recruitment is obtained from the male multi-stanza group.

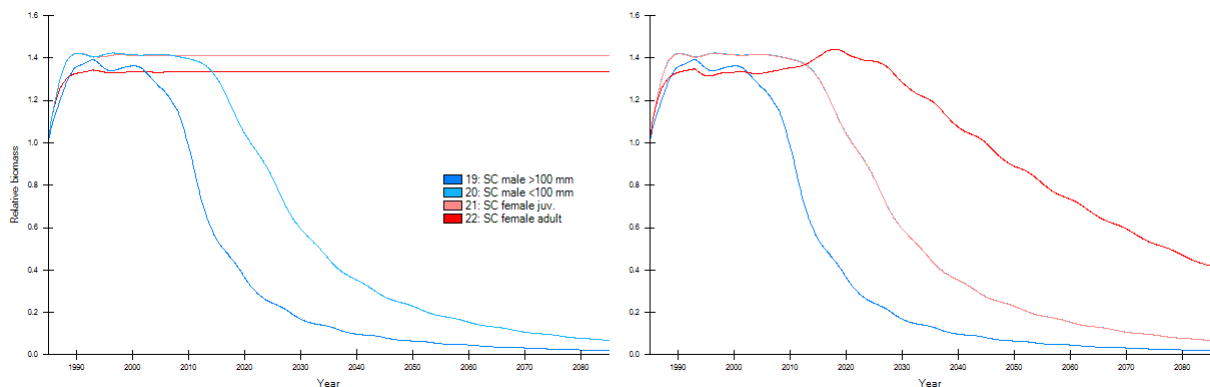


Figure 1 - Two Ecosim simulations for hypothetical fishing pressure on male snow crab, without (left) and with (right) recruitment of females obtained from the male multi-stanza group.

Configuring linked stanza recruitment

First, define the male and female multi-stanza configurations in the “Define groups” interface (Figure 2). In the EISA project, we explicitly modelled male and female snow crab as only male individuals with a carapace size over 100mm were fished.

	Group name	Color	Consumer	Producer	Detritus	Part of production from primary	Multi-stanza group name	Stanz (in mo)
14	Plank_pisc_fish		☒	☐	☐	0.000		
15	Pisc_fish		☒	☐	☐	0.000		
16	Small_benthivore_fish		☒	☐	☐	0.000		
17	Large_benthivore_fish		☒	☐	☐	0.000		
18	Shrimp		☒	☐	☐	0.000		
19	SC male >100 mm	Blue	☒	☐	☐	0.000	SC male	96
20	SC male <100 mm	Blue	☒	☐	☐	0.000	SC male	0
21	SC female juv.	Pink	☒	☐	☐	0.000	SC female	0
22	SC female adult	Red	☒	☐	☐	0.000	SC female	96
23	Red_king_crab		☒	☐	☐	0.000		
24	Large crabs		☒	☐	☐	0.000		
25	Pred_ivert		☒	☐	☐	0.000		
26	Depos_polychaete		☒	☐	☐	0.000		
27	Depos_echinoderm		☒	☐	☐	0.000		
28	Depos_other		☒	☐	☐	0.000		
29	Small_suspension		☒	☐	☐	0.000		
30	Large_suspension		☒	☐	☐	0.000		

Figure 2 - Define gender-split multi-stanza groups

Second, configure the two gender-split multi-stanza groups in the “Edit multi-stanza” interface (not shown here).

Third, link recruitment in the “Edit multi-stanza” interface (Figure 3). In the EISA project we linked recruitment of female snow crab to the male population as only the males were fished with indirect impacts on female crabs. A multi-stanza group whose recruitment is linked will have its recruitment calculations overwritten, and its default “Recruitment Power” parameter therefore becomes inactive in the “Edit multi-stanza” user interface (Figure 3).

Stanza group: 3: SC female

Curvature parameter, K (annual), from VBGF: 0.140

Recruitment power: 1.000

Recruitment stanza: 2: SC male

Relative biomass accumulation rate (BA/B): 0.170

Wmaturity/Winf: 0.0900

☐ Fixed fecundity

Ecosim

Forcing function for hatchery stocking: [dropdown]

Ecospace

☒ Recruit where spawned in Ecospace

Number at age 0 (n/km2/year): 0.00423

Group name	Age, start (months)	Leading biomass	Biomass (t/km²)	Tot. mort. (/year)	Leading Q/B	Consumption / biomass (/year)	Spawning (proportion)
21 SC female juv.	0	☐	0.000379	0.230	☐	3.383	0.000
22 SC female adult	96	☒	0.000112	0.450	☒	2.005	1.000

Figure 3 - Linking the recruitment of a multi-stanza configuration to another

References

- Christensen, V., Walters, C.J., 2004. Ecopath with Ecosim: methods, capabilities and limitations. *Ecol. Model.* 172, 109–139. <https://doi.org/10.1016/j.ecolmodel.2003.09.003>
- Heymans, J.J., Coll, M., Link, J.S., Mackinson, S., Steenbeek, J., Christensen, V., 2016. Best practice in Ecopath with Ecosim food-web models for ecosystem-based management. *Ecol. Model.* 331, 173–184. <https://doi.org/10.1016/j.ecolmodel.2015.12.007>
- Walters, C.J., Christensen, V., Walters, W., Rose, K.A., 2010. Representation of multistanza life histories in Ecospace models for spatial organization of ecosystem trophic interaction patterns. *Bull. Mar. Sci.* 86, 439–459.